

ARM-450 — Orthographic Three-View Drawings

12 sheets · third-angle projection · hidden detail shown dashed

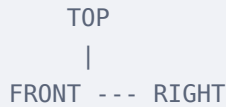


Diagram illustrating the layout of the three-view drawing: TOP view is positioned above the FRONT view, and the RIGHT view is positioned to the right of the FRONT view, separated by dashed lines. The text 'FRONT --- RIGHT' is shown at the bottom.

Every sheet carries FRONT, TOP and RIGHT views with overall dimensions, centre lines through the FRONT view, and a table of every cylindrical feature the STEP file actually contains — diameter, count, axis and depth.

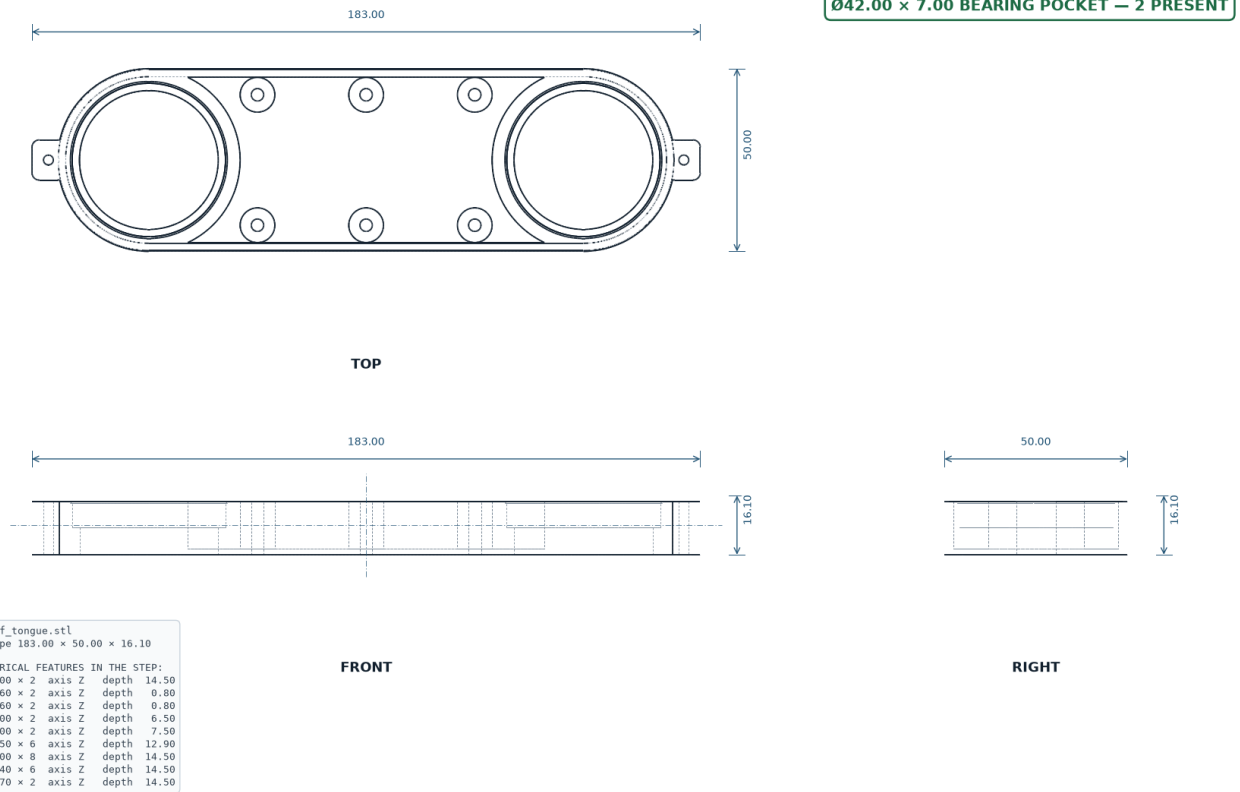
That table is the point. It is generated by reading the B-rep, so a sheet cannot claim a feature the part does not have. Three sheets carry a red banner because the audit found their bearing seat missing; see ARM450_AUDIT.pdf for the cause and the fix.

Hidden lines matter here. The bearing pockets, blind insert holes and internal webs are precisely the features invisible from outside, and they are what the audit is about. Solid lines are visible edges; dashed grey lines are edges behind material.

Views are of the part as modelled, not as oriented on the print bed. Print orientation is on the isometric sheets in ARM450_DRAWINGS_ISO.pdf.

1. Link half — tongue side

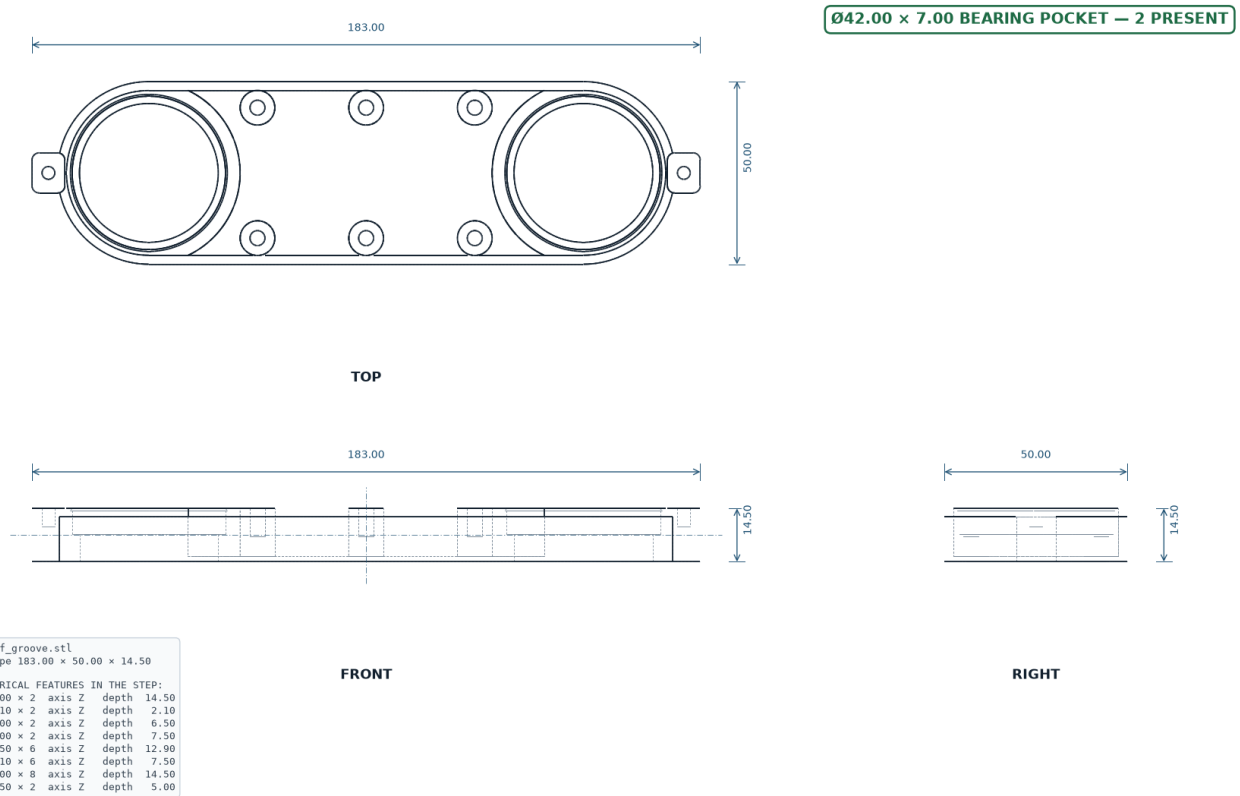
LINK HALF — TONGUE SIDE — third-angle projection



2 off. Carries the blocking defect: no Ø42 bearing pocket. The FRONT view shows the boss wall running straight from web to seam face with no counterbore step.

2. Link half — groove side

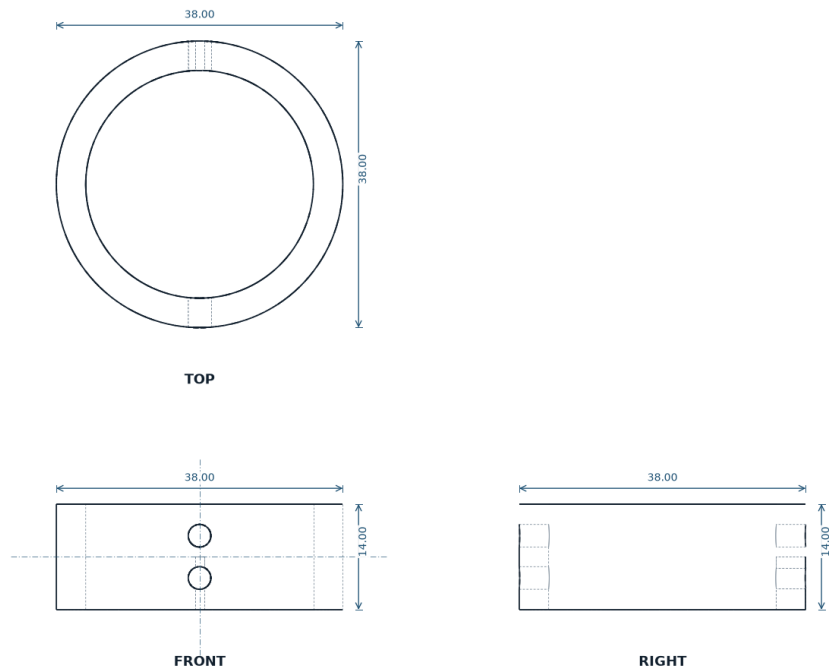
LINK HALF — GROOVE SIDE — third-angle projection



2 off. Same defect. The TOP view shows the six Ø4.10 insert holes and the two tip ears; the dashed circles in FRONT are the blind insert bottoms.

3. Shaft clamp

SHAFT CLAMP (SPLIT) — third-angle projection



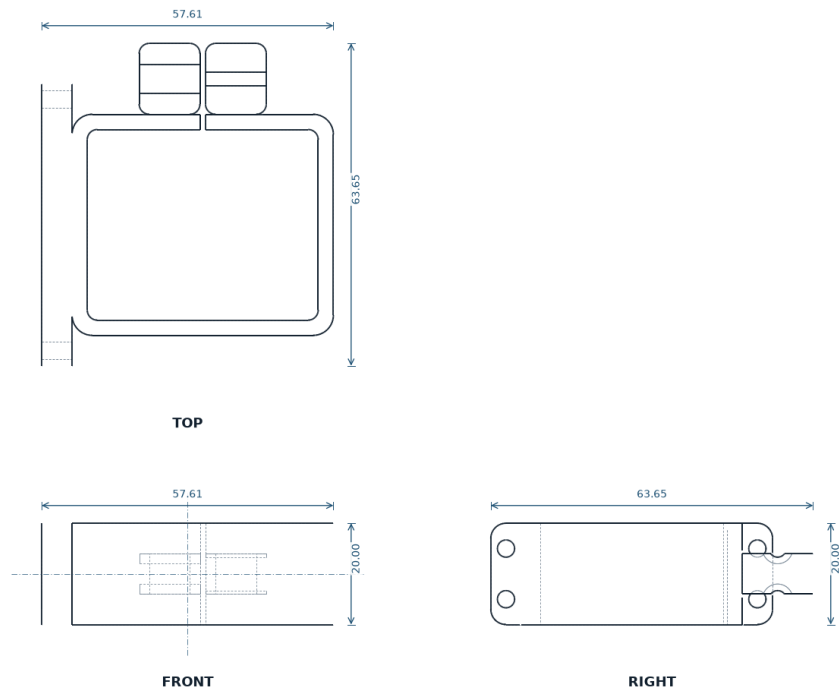
```
shaft_clamp.stl
envelope 38.00 x 38.00 x 14.00

CYLINDRICAL FEATURES IN THE STEP:
Ø 38.00 x 1 axis Z depth 14.00
Ø 30.20 x 1 axis Z depth 14.00
Ø 3.00 x 4 axis Y depth 3.97
```

2 off. TOP shows the split slot at the top of the ring; the two $\text{Ø}3.00$ roll-pin holes appear as dashed bores in FRONT and as solid circles where they break the surface.

4. Servo collar

SERVO COLLAR — third-angle projection



```
servo_collar.stl
envelope 57.61 x 63.65 x 20.00

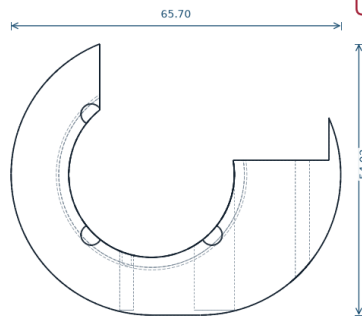
CYLINDRICAL FEATURES IN THE STEP:
Ø 8.00 x 4 axis Z depth 20.00
Ø 6.00 x 6 axis X depth 6.00
Ø 4.00 x 12 axis Z depth 20.00
Ø 3.40 x 6 axis X depth 6.00
```

2 off. The dashed rectangle in TOP is the servo pocket; the six Ø3.40 fixings appear in RIGHT.

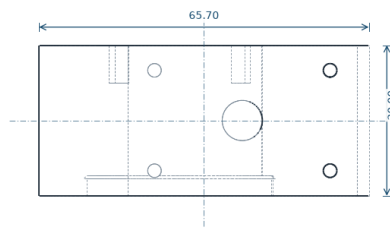
5. J4 roll housing

J4 ROLL HOUSING — third-angle projection

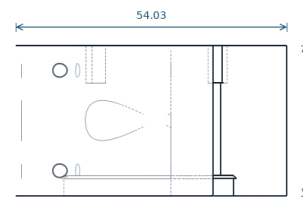
NO Ø42.00 BEARING POCKET — see ARM450_AUDIT.pdf



TOP



FRONT



RIGHT

```
wrist_j4_housing.stl
envelope 65.70 x 54.03 x 30.00

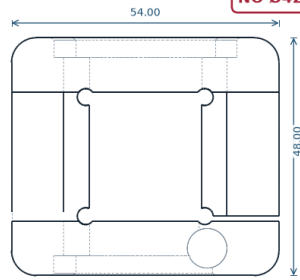
CYLINDRICAL FEATURES IN THE STEP:
Ø 56.00 x 2 axis Z depth 30.00
Ø 37.00 x 1 axis Z depth 3.50
Ø 33.00 x 1 axis Z depth 26.00
Ø 8.00 x 1 axis Y depth 29.88
Ø 4.10 x 3 axis Z depth 7.50
Ø 2.70 x 4 axis Y depth 11.77
```

1 off. Ø42.00 pocket and Ø38.00 through bore both present and coaxial to 0 µm.

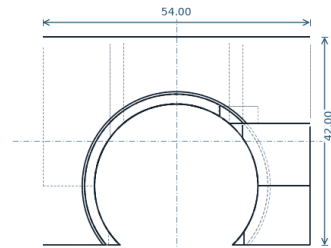
6. J5 pitch yoke

J5 PITCH YOKE — third-angle projection

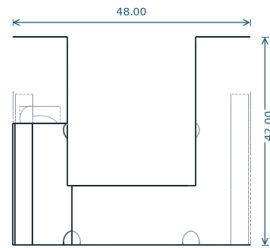
NO Ø42.00 BEARING POCKET — see ARM450_AUDIT.pdf



TOP



FRONT



RIGHT

```
wrist_j5_yoke.stl
envelope 54.00 x 48.00 x 42.00

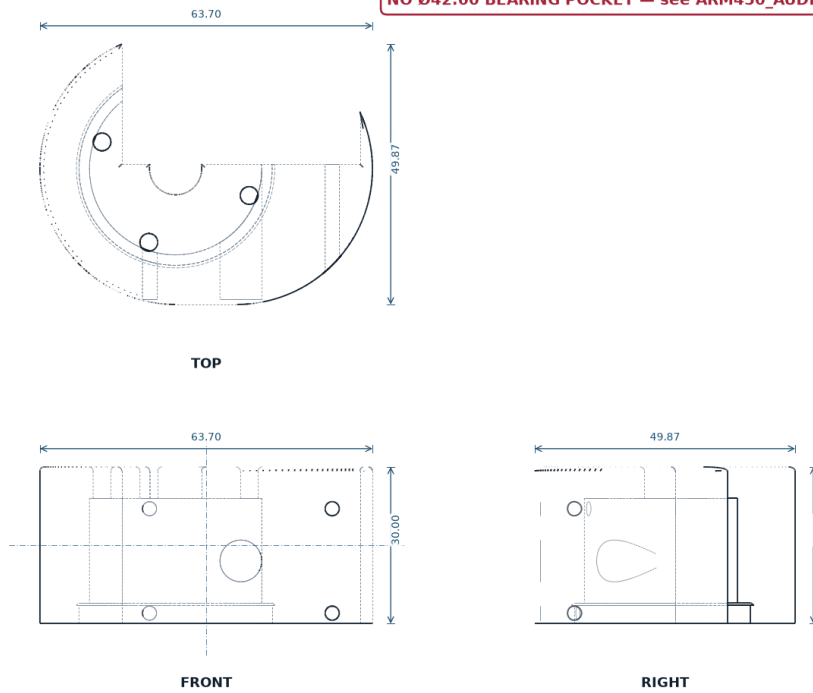
CYLINDRICAL FEATURES IN THE STEP:
Ø 37.00 x 2 axis Y depth 3.50
Ø 33.00 x 1 axis Y depth 40.02
Ø 10.00 x 4 axis Z depth 42.00
Ø 8.00 x 1 axis Z depth 3.60
Ø 3.40 x 0 axis Z depth 2.84
```

1 off. Two Ø42.00 pockets, one per cheek, on a single Ø38.00 bore. Measured bearing-face spacing is 38.02, not the 40.00 quoted elsewhere.

7. J6 output / tool flange

J6 OUTPUT / TOOL FLANGE — third-angle projection

NO Ø42.00 BEARING POCKET — see ARM450_AUDIT.pdf



```
wrist_j6_output.stl
envelope 63.70 x 49.87 x 30.00

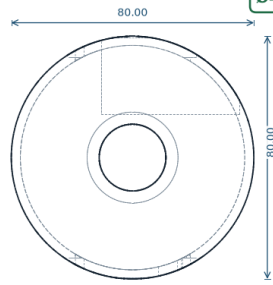
CYLINDRICAL FEATURES IN THE STEP:
Ø 52.00 x 2 axis Z depth 29.20
Ø 37.00 x 1 axis Z depth 3.50
Ø 33.00 x 1 axis Z depth 20.00
Ø 10.00 x 1 axis Z depth 5.20
Ø 8.00 x 1 axis Y depth 25.88
Ø 3.40 x 3 axis Z depth 5.20
Ø 2.70 x 4 axis Y depth 7.62
```

1 off. Ø10.00 centre pilot now present. Only three of four tool-face bolt holes exist — the fourth falls in the servo pocket opening.

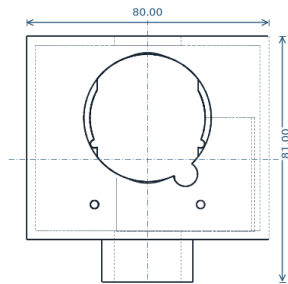
8. J1 turret

J1 TURRET — third-angle projection

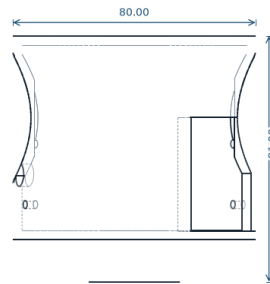
Ø42.00 × 7.00 BEARING POCKET — 2 PRESENT



TOP



FRONT



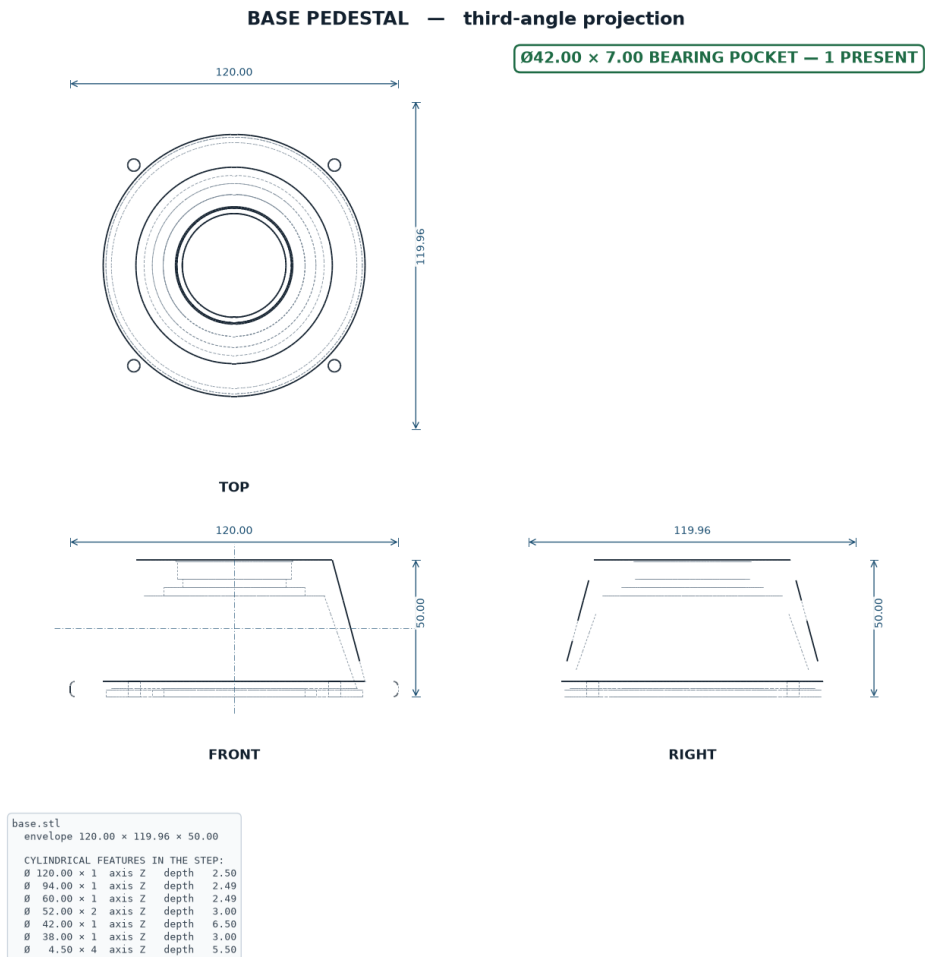
RIGHT

```
turret_j1.stl
envelope 80.00 x 80.00 x 81.00

CYLINDRICAL FEATURES IN THE STEP:
Ø 80.00 x 1 axis Z depth 67.00
Ø 74.00 x 1 axis Z depth 61.00
Ø 42.00 x 2 axis Y depth 6.50
Ø 38.00 x 2 axis Y depth 1.26
Ø 30.00 x 1 axis Z depth 14.00
Ø 22.00 x 2 axis Z depth 17.00
Ø 8.00 x 1 axis Y depth 5.88
Ø 2.70 x 6 axis Y depth 4.76
```

1 off. Two Ø42.00 pockets on axis Y. No horn-circle holes are present — see audit finding 6.

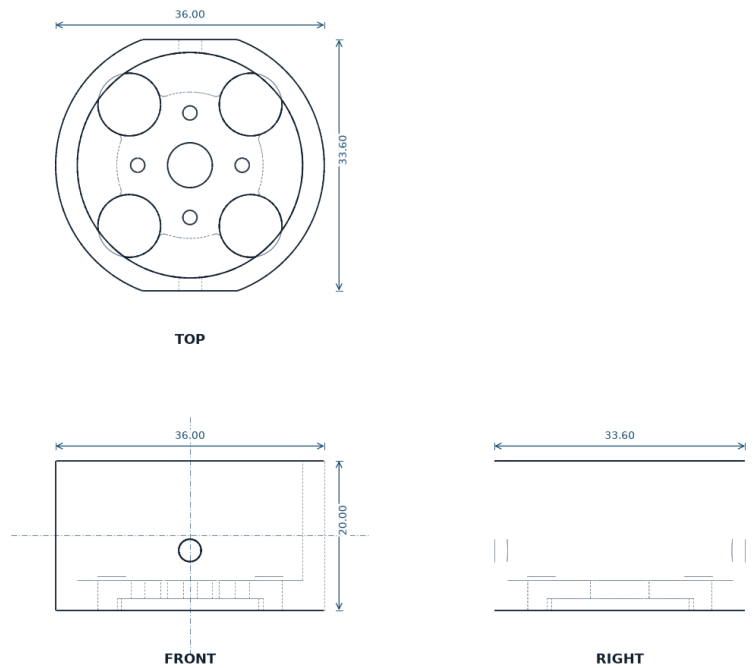
9. Base pedestal



1 off. Blocking defect: no Ø42 J1 bearing seat. Both pedestal walls end in a plain flat rim in the FRONT view.

10. Servo horn adapter

SERVO HORN ADAPTER — third-angle projection



```

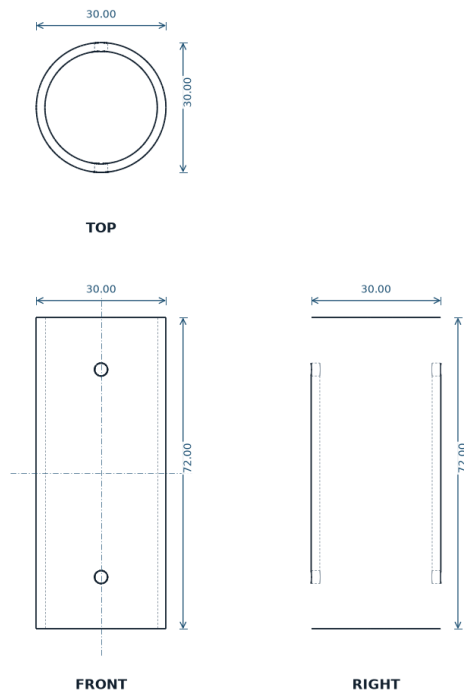
horn_adapter.stl
envelope 36.00 x 33.60 x 20.00

CYLINDRICAL FEATURES IN THE STEP:
Ø 36.00 x 1 axis Z depth 20.00
Ø 30.20 x 1 axis Z depth 16.00
Ø 19.60 x 1 axis Z depth 1.60
Ø 8.40 x 4 axis Z depth 4.50
Ø 6.00 x 1 axis Z depth 2.40
Ø 3.00 x 2 axis Y depth 1.77
    
```

6 off. The part that closes the torque path. Until it existed the servo horn had nothing to drive: the old solid shaft carried its own horn shoulder, and that disappeared when the shaft became a bought aluminium tube. The TOP view shows the four Ø1.90 horn holes on the Ø14 bolt circle and the scalloped face between them; the dashed bore in FRONT is the Ø30.20 slip fit over the tube, and the cross hole is the roll pin.

11. Joint shaft — aluminium tube

JOINT SHAFT — AL TUBE — third-angle projection



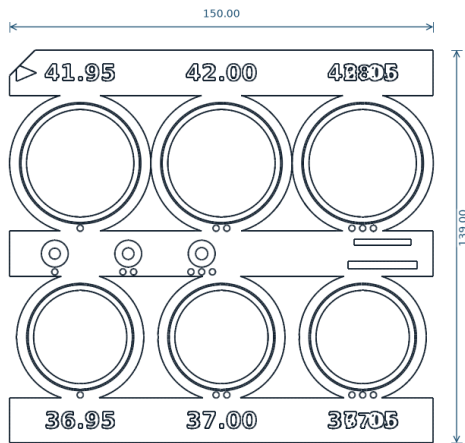
```
shaft_tube.stl
envelope 30.00 x 30.00 x 72.00

CYLINDRICAL FEATURES IN THE STEP:
Ø 30.00 x 1 axis Z depth 72.00
Ø 26.00 x 1 axis Z depth 72.00
Ø 3.00 x 4 axis Y depth 2.09
```

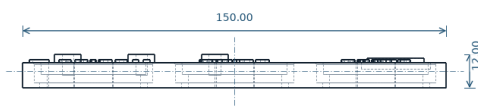
6 off, bought. Ø30.00 OD / Ø26.00 bore, coaxial to 0 µm, two Ø3.00 cross holes 12.00 from each end.

12. Fit-test coupon

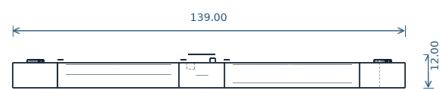
FIT-TEST COUPON — third-angle projection



TOP



FRONT



RIGHT

```
fit_coupon.stl
envelope 150.00 x 139.00 x 12.00

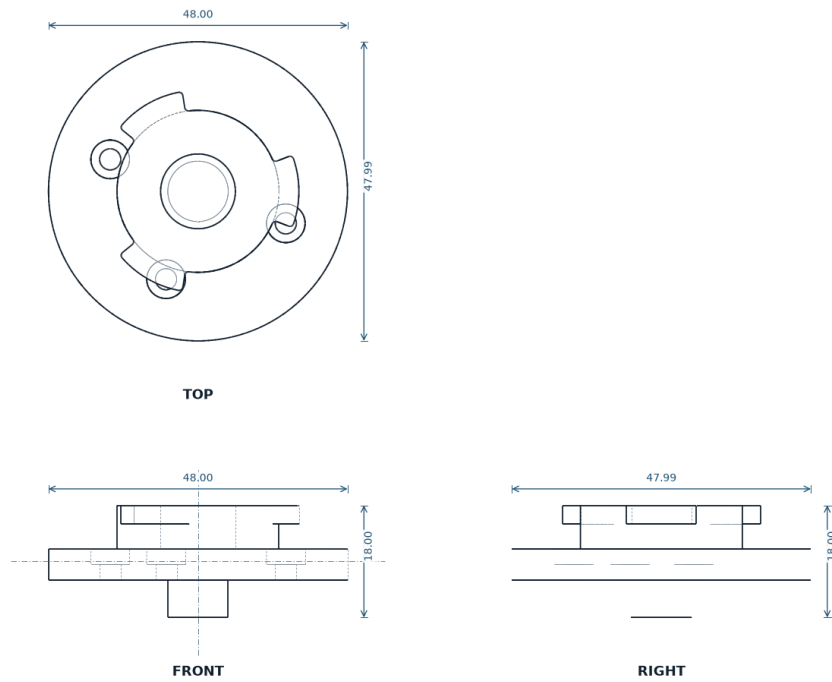
CYLINDRICAL FEATURES IN THE STEP:
Ø 50.00 x 3 axis Z depth 9.00
Ø 45.00 x 3 axis Z depth 9.00
Ø 42.05 x 1 axis Z depth 6.50
Ø 42.00 x 1 axis Z depth 6.50
Ø 41.95 x 1 axis Z depth 6.50
Ø 38.00 x 3 axis Z depth 2.00
Ø 37.05 x 1 axis Z depth 3.50
Ø 37.00 x 1 axis Z depth 3.50
Ø 36.95 x 1 axis Z depth 3.50
Ø 33.00 x 3 axis Z depth 5.00
Ø 9.50 x 3 axis Z depth 3.00
```

1 off. The three test pockets $\varnothing 41.95$ / $\varnothing 42.00$ / $\varnothing 42.05$ are visible in TOP and as dashed bottoms in FRONT.

End effectors

13. Tool adapter

TOOL ADAPTER — ARM SIDE OF THE BAYONET — third-angle projection

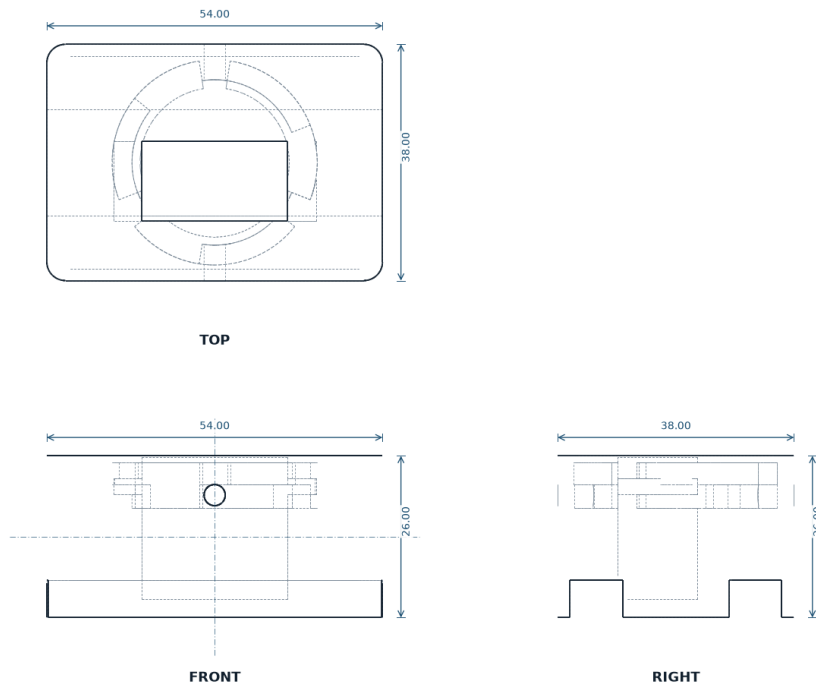


tool_adapter.stl				
envelope 48.00 x 47.99 x 18.00				
CYLINDRICAL FEATURES IN THE STEP:				
Ø 48.00	x 1	axis Z	depth	5.00
Ø 32.40	x 1	axis Z	depth	3.00
Ø 26.00	x 1	axis Z	depth	7.00
Ø 12.00	x 1	axis Z	depth	7.00
Ø 9.70	x 1	axis Z	depth	6.00
Ø 6.20	x 3	axis Z	depth	2.40
Ø 3.40	x 3	axis Z	depth	2.60

1 off. The three M3 counterbores and the Ø9.70 pilot spigot show in FRONT; the bayonet lugs in TOP.

14. Gripper body

GRIPPER BODY — SG90 PARALLEL JAW — third-angle projection



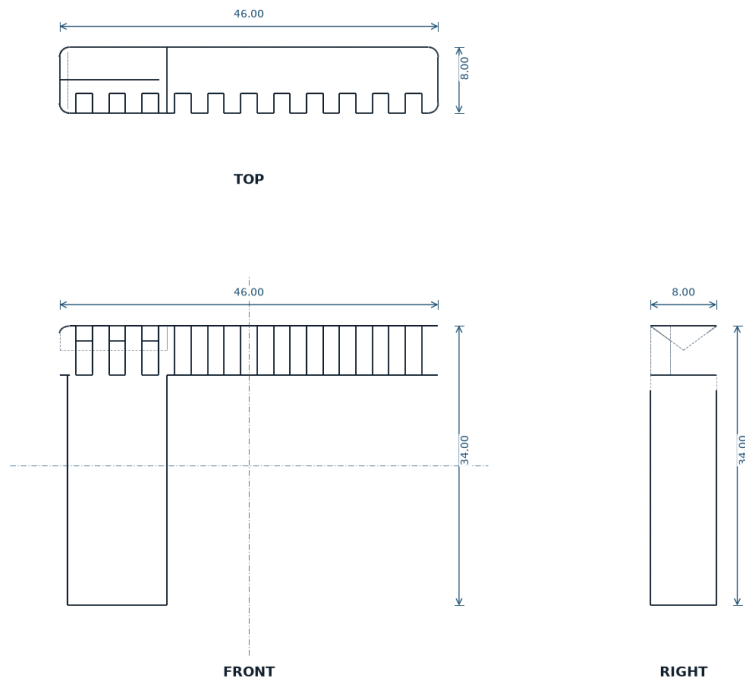
```
tool_gripper.stl
envelope 54.00 x 38.00 x 26.00

CYLINDRICAL FEATURES IN THE STEP:
Ø 33.00 x 1 axis Z depth 7.31
Ø 26.60 x 1 axis Z depth 7.30
Ø 24.00 x 1 axis Z depth 0.01
Ø 6.00 x 4 axis Z depth 26.00
Ø 3.40 x 2 axis Y depth 5.81
```

1 off. The SG90 pocket and both rack channels are the dashed detail in FRONT.

15. Gripper jaw

GRIPPER JAW (PRINT 2, ONE MIRRORED) — third-angle projection

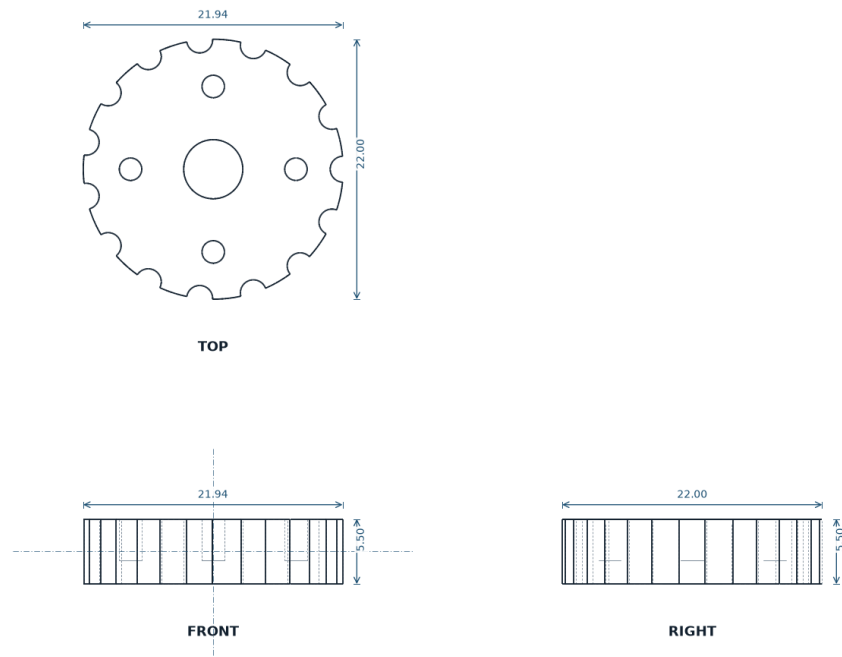


gripper_jaw.stl
envelope 46.00 x 8.00 x 34.00
CYLINDRICAL FEATURES IN THE STEP:
Ø 2.40 x 4 axis Z depth 5.93

2 off, second mirrored. RIGHT shows the rack teeth against the V-groove in the finger.

16. Gripper pinion

GRIPPER PINION — third-angle projection



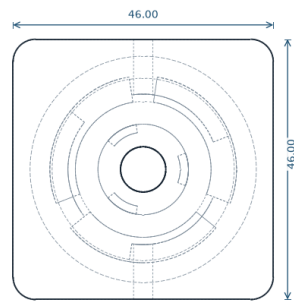
```
gripper_pinion.stl
envelope 21.94 x 22.00 x 5.50

CYLINDRICAL FEATURES IN THE STEP:
Ø 22.00 x 1 axis Z depth 5.50
Ø 5.00 x 1 axis Z depth 5.50
Ø 2.20 x 15 axis Z depth 5.50
```

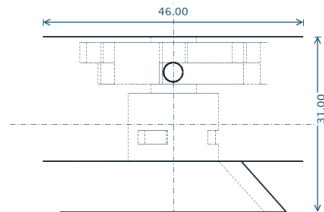
1 off, 100 % infill.

17. MASCOT docking probe

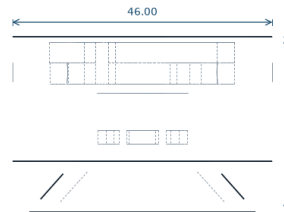
MASCOT DOCKING PROBE — third-angle projection



TOP



FRONT



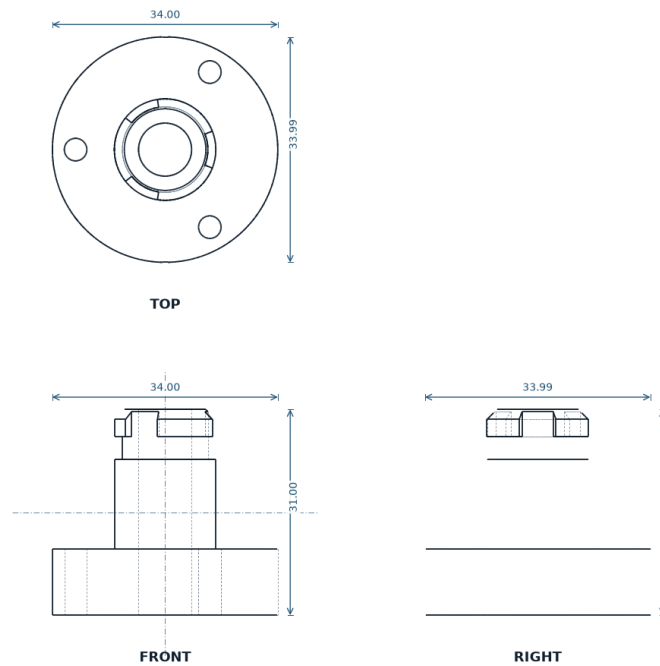
RIGHT

tool_dock.stl
envelope 46.00 x 46.00 x 31.00
CYLINDRICAL FEATURES IN THE STEP:
Ø 33.00 x 1 axis Z depth 7.31
Ø 26.60 x 1 axis Z depth 7.30
Ø 24.00 x 1 axis Z depth 0.01
Ø 16.00 x 1 axis Z depth 12.00
Ø 13.10 x 1 axis Z depth 2.40
Ø 8.00 x 6 axis Z depth 22.00
Ø 3.40 x 2 axis Y depth 9.81

1 off. FRONT carries the whole story: the Ø34 mouth, the taper to the Ø16 throat, and the latch lugs inside it.

18. Docking target port

DOCKING TARGET PORT (BENCH TEST) — third-angle projection



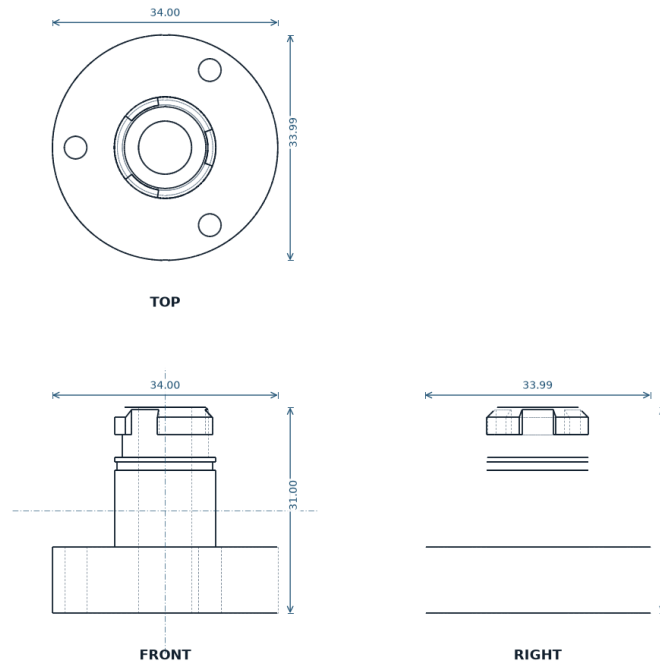
```
dock_target.stl
envelope 34.00 x 33.99 x 31.00

CYLINDRICAL FEATURES IN THE STEP:
Ø 34.00 x 1 axis Z depth 10.00
Ø 15.30 x 2 axis Z depth 13.50
Ø 12.90 x 1 axis Z depth 7.20
Ø 8.00 x 1 axis Z depth 31.00
Ø 3.40 x 3 axis Z depth 10.00
```

1 off. The capture groove and the three entry slots are the dashed detail in FRONT.

19. Docking target port — sealed

DOCKING TARGET PORT — SEALED, FOR FLUID — third-angle projection



```
dock_target_sealed.stl
envelope 34.00 x 33.99 x 31.00

CYLINDRICAL FEATURES IN THE STEP:
Ø 34.00 x 1 axis Z depth 10.00
Ø 15.30 x 3 axis Z depth 11.55
Ø 14.40 x 1 axis Z depth 1.30
Ø 12.90 x 1 axis Z depth 7.20
Ø 8.00 x 1 axis Z depth 31.00
Ø 3.40 x 3 axis Z depth 10.00
```

1 off, an ALTERNATIVE to dock_target. FRONT carries the O-ring groove in the spigot.

Regenerating

```
cd ~/ros2_ws/arm450_design
python3 draw_3view.py # rewrites figures/3view/*.png
python3 build_pdf.py DRAWINGS_3VIEW.md
```

orthodraw.py does the projection and hidden-line removal; draw_3view.py lays out the sheets and pulls the feature table from audit_parts.py. Nothing on these sheets is typed by hand.